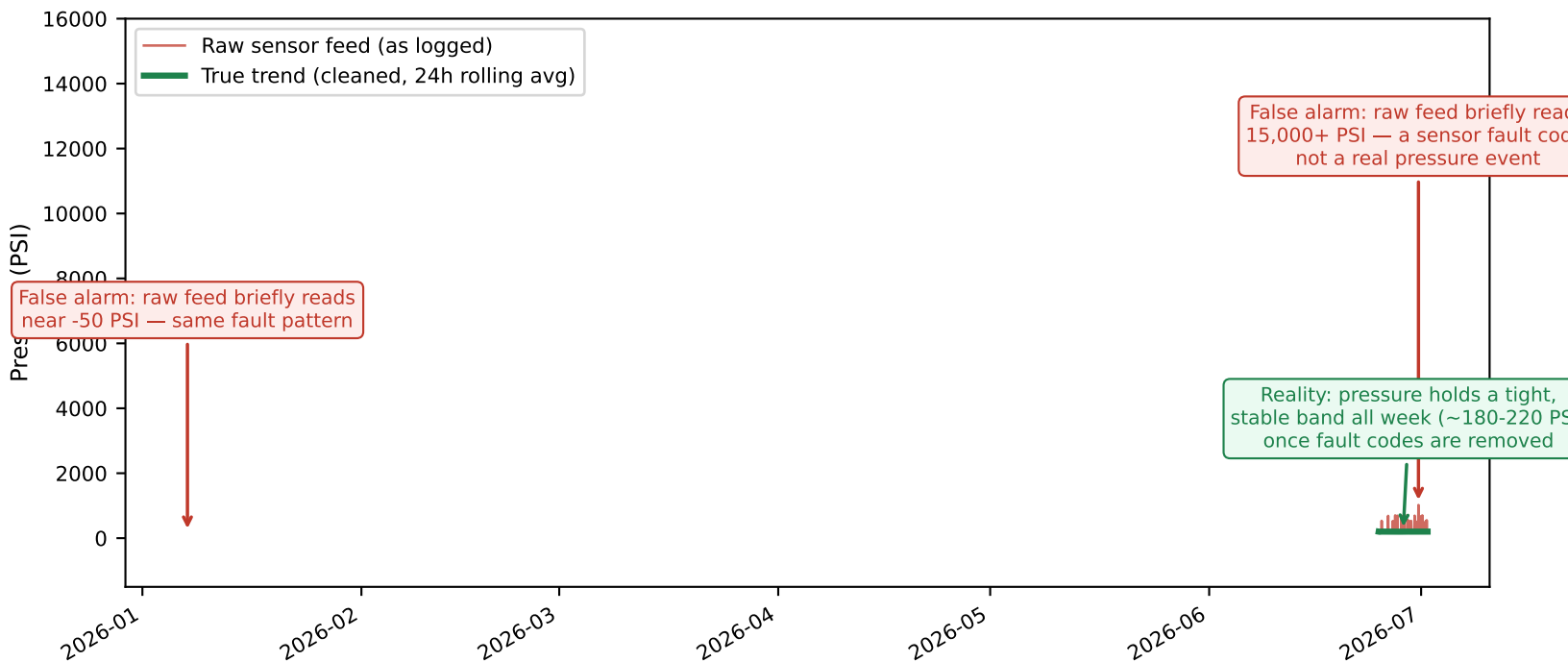


Prepared for: Plant Manager

Period reviewed: 25 Jun – 1 Jul 2026

Line Pressure (PSI): What the Raw Feed Showed vs. What Was Actually Happening**CONTEXT — What was wrong with the raw data**

This week's raw sensor feed made the plant look dangerously erratic — pressure appeared to swing from below zero to over 15,000 PSI within minutes. That reading is not physically possible on this line. It came from 75 individual sensor fault and drop-to-zero events scattered across the week, not from an actual process upset. Left unfiltered, a dashboard or alarm system reading this feed directly would be flooded with false alarms and would obscure any real fault if one occurred.

INSIGHT — What cleaning revealed

After removing the fault codes and rebuilding the true trend, plant pressure was in fact stable all week, holding a tight band around 200 PSI with only normal minor variation. The hidden finding is not an operational instability — it is a sensor reliability problem, and it is not evenly spread: Zone_North and Zone_South together account for roughly 45% of all fault/dropout events (17 and 18 of 75), despite the plant having five zones of similar size. Two zones are generating almost half of all bad readings.

ACTION — What we recommend

Recommend a targeted sensor inspection and recalibration in Zone_North and Zone_South within the next maintenance window, prioritized ahead of the other three zones. In parallel, add a simple range check (reject readings outside 50-400 PSI and 10-200 C) at the point of ingestion so fault codes never reach live dashboards or trigger alarms again. This is a low-cost fix that removes false alarms without masking a real pressure event if one occurs.